
SOFTWARE REQUIREMENTS SPECIFICATION

for

BookStack

Version 1.0

Prepared by :

Cade Besaw

Matias Bustillo

Cara Belew

Leonel Giacobbe

Mississippi State University

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1 Introduction

1.1 Purpose

The software requirements specification for this e-commerce website intends to describe the requirements, features, and design plans for the application. This document also serves as a tool for both the team and the customer, to communicate the plans, requirements, and ramifications for the website.

1.2 Document Conventions

Bold Text = Section titles *Italic Text* = Definitions or emphasis on a detail **Priority levels (High, Medium, Low)** is categorized by (H,M,L) respectively.

Acronyms:

- SRS: Software Requirements Specification

1.3 Intended Audience and Reading

This document is intended for multiple different parties, including the developers of the team, the customers for the website, and anyone else interested in learning about the e-commerce website. Section 1 is an overview of the system, going over the purpose of the software and what to look for in the document, Section 2 is an overall description of the product, going over the different functions, user classes, characteristics, operating environment, and the design and implementation constraints. Section 3 gets more specific and detailed with the functional requirements, while Section 4 goes over the nonfunctional requirements. Section 5 looks at any additional constraints and potential considerations for the website.

1.4 Product Scope

The software being developed has the goal of serving as an online marketplace for books. Users can buy and sell new and used books, having the opportunity to recycle material and obtain books at a lower price.

The goal of the software is to provide a safe and secure environment for the activities mentioned above.

The company can profit in multiple ways:

- Charging a percentage of every sale

- Subscription based model
- Ads placed on website
- Promotions paid by users to place their posts at the top of feeds

For this implementation, the following areas will be inside the scope of the project:

- User management
- Website browsing
- Visiting profiles
- Making purchases
- Publishing items for sale

On the contrary, here's what will be outside the scope of the project:

- Email verification
- Delivery handling (will be simulated)
- Conversations between users
- Negotiation of prices

1.5 References

TypeScript: <https://www.typescriptlang.org/>

Python: <https://www.python.org/>

PostgreSQL: <https://www.postgresql.org/>

Spring Boot: <https://spring.io/projects/spring-boot>

2 Overall Description

2.1 Product Perspective

This is a new, standalone project designed to function as a book buying and selling marketplace. The system will follow a traditional frontend and backend pairing, with a database being used for data persistence. The front end allows Buyers, Sellers, and Admins to interact with the system. Admins will be given administrative power and the ability to delete users if needed. Buyers and sellers will be able to look through the items, choose items and add them to the cart, purchase items, and be able to sell their own items / put them on the marketplace. In the backend, the website will have a centralized server that handles user authentication and the types of roles, logging and monitoring of user actions, product listings, orders, returns, approvals, and payment processing. The database layer stores user accounts / roles, product listings, activity logs for admins, and orders.

2.2 Product Functions

2.2.1 User Handling

- Create accounts: users should be able to input a combination of email and password to create an account.
- Log in: users should be able to log in and use the website with their accounts if they enter a correct combination of email and password
- Edit account info: users should be able to change their account information, such as name, address, username, etc.

2.2.2 Browsing

- Search items by title: users will be able to look up books by their title. Exact matches will not be necessary.
- Search items by author: users will be able to look up books by their author.
- Filter items by price range: users can limit search results based on a maximum allowed price.
- Filter items by year: users can limit search results based on a range of years.

- Visit another user's profile: users will be able to see other users' public information, such as username, name, and for-sale items in their profile.

2.2.3 Purchasing

- Add items to cart: users will be able to click on items and add them to their cart, then being able to purchase multiple items at once.
- Empty or remove items from cart: users will be able to remove one or all items at once from their cart.
- Check out: users will be able to proceed with a purchase once at least one item is present in their cart

2.2.4 Administration

- View audit logs: administrators will be allowed to see the logs produced by the program.
- Ban users: administrators will be allowed to remove the accounts of users found to be in violation of the rules of the service.

2.3 User Classes and Characteristics

2.3.1 Regular users (buyers and sellers)

These users will use the website on a semi-frequent basis. They will mostly use the functions related to the buying and selling of books, and occasionally will update their account information. They can see other users' profiles but will have access to a more restricted view compared to the view that administrators have.

2.3.2 Administrators

The administrators are responsible for moderating the behavior of users and managing them. They have the authority to ban and delete accounts when they consider appropriate. They have access to overarching information regarding the website, such as a feed of all recent transactions, recent actions taken by a user, and can manage claims. They do not have access to users' sensible information.

2.4 Operating Environment

The software will be deployed using Docker Compose in a group member's home server. The components will be containerized using Docker. The main hosting platform to handle the incoming requests for the website will be Spring Boot. Cloud SQL (PostgreSQL) will be used for databases. The project will be made publicly accessible with the use of LocalTunnel.

2.5 Design and Implementation Constraints

For security considerations, we will not develop the option to chat between users. Regulatory policies would mandate moderation of these chats and, given that the developers are college students, that is not something we can do at this point.

Considering our status as students and given that this project is being developed for a class, we will not be employing any paid-for software or service besides the hosting technologies.

3 System Features

3.1 Log in / Sign Up (High Priority)

3.1.1 Description and priority

This feature enables users to create accounts, log in, log out, and access the system functionalities based on their assigned role. This is a high priority feature.

3.1.2 Stimulus/Response Sequences

- Stimulus (1): The user enters valid login credentials.
- Response (1): The system allows the user to log in and grants access to the website's features based on their role.
- Stimulus (2): User attempts to access a feature delegated to admin only.
- Response (2): System denies access to user and prompts an authorization error.
- Stimulus (3): User logs out
- Response (3): System terminates session, logs out user, and returns to login page.
- Stimulus (4): User enters invalid login credentials.
- Response (4): The system informs this to the user and prompts them to enter the credentials again.
- Stimulus (5): User tries to create an account with an email or username that is already in use.
- Response (5): The system informs this to the user and prompts them to enter another email or username.
- Stimulus (6): The user enters valid login credentials.
- Response (6): The system allows the user to log in and grants access to the website's features based on their role.

3.1.3 Functional Requirements

- REQ-1: The system shall display a login/ sign-up form with text boxes for username/ email and password, as well as a submit button.
- REQ-2: The system shall display an error message whenever a username and password combination is incorrect.
- REQ-3: The system shall display an error message whenever a username or email that tries to be used to create an account is already present in the database.
- REQ-4: If the inputted password during account creation is less than 6 characters long, the system shall inform the user of this.
- REQ-5: The system shall not display the context of the user inputs in the password field and instead display circles to correspond for every character.
- REQ-6: If all the information in the sign-up page is valid, the system shall create an entry in the database containing all the user information after the user clicks enter.
- REQ-7: If all the information is correct in the sign-in page, the system shall log the user in.

3.2 Search Marketplace (High Priority)

3.2.1 Description and priority

This feature allows customers to search by title and author, and filter by price range and release date. From here, customer can select books to get additional information and choose to add to cart.

3.2.2 Stimulus/Response Sequences

- Stimulus (1): User searches for a book using the title or author.
- Response (1): System displays all matching book(s).
- Stimulus (2): User selects a price range limit for books.
- Response (2): queried books get filtered by the price range and the user is only shown those that fall between the range.
- Stimulus (3): User selects a year range for books.
- Response (3): queried books get filtered by the release year and the user is only shown those that fall between the range.

3.2.3 Functional Requirements

- REQ-8: The system shall display a horizontal search bar at the top of the website.
- REQ-9: The system shall allow the user to input text onto the search bar after it is clicked.
- REQ-10: The system shall display three buttons under the search bar: "Year", "Author", "Price Range".
- REQ-11: After the user clicks on the "Year" button, the system shall display a text box for users to input a year.
- REQ-12: The system shall limit the input in the text box corresponding to the "Year" filter to numbers only, and should be a number that is not in the future.
- REQ-13: After the user clicks on the "Author" button, the system shall display a text box for users to input an Authors name.
- REQ-14: The system shall limit the input in the text box corresponding to the "Author" filter to letters only.
- REQ-15: After the user clicks on the "Price Range" button, the system shall display a slider bar that represents the maximum price. The lower end shall always be zero, and sliding the bar to the right will increase the allowed maximum price.
- REQ-16: After the user enters text into the search box and, optionally, fills in the filter fields, the system shall query the database according to the provided information.

3.3 Product Details (Medium Priority)

3.3.1 Description and priority

This feature allows the user to see more details about a book in the marketplace, including stock, book cover, condition, edition, year, author, genre, and short description.

3.3.2 Stimulus/Response Sequences

Precondition: User is viewing a list of books

- Stimulus (1): User clicks a book
- Response (1): System obtains the book ID
- Response (1): Queries for relevant metadata associated with the captured ID
- Response (1): System loads product details page

- Response (1): System Displays the Static Data: Book Title, Author, Genre, Edition, Condition, Published Date, Short Description
- Response (1): System Displays Media: High Resolution Cover Image

3.3.3 Functional Requirements

- REQ-17: System shall make books clickable.
- REQ-18: Upon the user clicking on a book, system shall redirect them to another page, where more detailed information about that book shall be displayed.

3.4 Administrative Capabilities (High Priority)

3.4.1 Description and priority

This feature allows admin to delete or block user accounts and product listings, approve accounts and book postings, as well as a number of administrative operations. This is a safety measure to monitor platform activity.

3.4.2 Stimulus / Response Sequences

- Stimulus (1): Admin clicks on a product listing:
- Response (1): Server displays the usual product information, plus an extra button to delete the listing.
- Stimulus (2): Admin clicks on the button to delete the listing.
- Stimulus (2): Listing is removed from user's account and from any carts that contained that listing.
- Stimulus (3): Admin visits audit log page.
- Response (3): Admin can see a complete log of most of the actions taken in the page, including recent purchases and other user activity, for moderation purposes.
- Stimulus (4): Admin visits the "pending approval" page.
- Response (4): Admin can see a list of all books and accounts pending approval. Book and account details are shown, to be examined by the admin and take a decision.

3.4.3 Functional Requirements

- REQ-19: If a user is logged in and they possess the "admin" role, the system shall display an "Admin Log" button on the top right of the home page.
- REQ-20: Upon an admin clicking on that button, they shall be redirected to the main admin logs page.
- REQ-21: While in the admin logs page, the system shall display a canonical log of all the activity taken by the users of the website, being careful to censor any private information for the sake of privacy.

3.5 User Account Page (Medium Priority)

3.5.1 Description and priority

This feature allows users to view/edit personal account information such as: profile picture, displayed name, account username, password, address. The user will also be able to view/edit their current listings.

3.5.2 Stimulus/Response Sequences

- Stimulus (1): User clicks edit button in the box containing personal account information.
- Response (1): System displays a pop-up containing text boxes for the user's information to be edited.
- Stimulus (2): User clicks edit button in any of their active listings.
- Response (2): System displays a pop-up menu enabling the users to edit the listing.
- Stimulus (3): User clicks delete button in any of their active listings.
- Response (3): System removes listing from the database, making it no longer visible to other users and removing it from any cart.
- Stimulus (4): User clicks button to delete account.
- Response (4): System displays a pop-up menu asking for confirmation. Upon confirmation, user information is deleted from the database, including their active listings.

3.5.3 Functional Requirements

- REQ-22: Upon users clicking on their profile picture, the system shall display a popup menu, prompting them to open their library to upload a picture.

- REQ-23: Upon the user uploading a new profile picture, the system shall replace the old one with the new one in the database.
- REQ-24: The system shall display the following buttons: "Change your username", "Change your password", "Change your email" and "Delete your account"
- REQ-25: Upon users clicking on a "Change your username", "Change your email" and "Change your password" button, the system shall display a popup menu with a text box for user input and a button to confirm the changes.
- REQ-26: Upon users clicking the "confirm" button, the system shall perform a number of validity checks on the new fields.
- REQ-27: Upon users clicking on the "Delete your account" button, the system shall display a pop-up menu with a message resembling "Are you sure you want to delete your account?" with "YES" and "NO" buttons.
- REQ-28: Upon the user clicking the "YES" button, all their information shall be deleted from the database, including their listings.
- REQ-29: Upon the user clicking the "NO" button, the system shall redirect them back to the "Edit your account information" page.

3.6 Cart and checkout Page (High Priority)

3.6.1 Description and priority

This feature allows users to view and modify their cart. They will be able to see all the books they have added, as well as total price of all the items.

3.6.2 Stimulus/Response Sequences

- Stimulus (1): User clicks on cart button.
- Response (1): System navigates user to cart page and displays relevant information.
- Stimulus (2): User elects to remove a book from their cart.
- Response (2): The book will no longer show up in the cart page and the new total will be calculated.
- Stimulus (3): User checks out the cart.
- Response (3): System removes listing(s) from the database, making it no longer visible to other users. It empties the user's cart and simulates a purchase.
- Stimulus (4): User clicks button to empty cart.

- Response (4): All books are removed from the cart, and the total gets reset to zero.

3.6.3 Functional Requirements

- REQ-30: The system shall display a button with a cart icon on the right of the menu taskbar.
- REQ-31: Upon users clicking the cart button, the system shall redirect them to the cart page.
- REQ-32: In the cart page, the system shall display the items currently in their cart, the subtotal, the total after tax (assumed to be 7 percent), and two buttons: a "CHECKOUT" button, and an "EMPTY CART" button.
- REQ-33: Upon users clicking on the "CHECKOUT" button, the system shall simulate a purchase using fictional credit card information
- REQ-34: Upon users clicking on the "EMPTY CART" button, all items shall be removed from their cart.
- REQ-35: Next to every item currently on a cart, the system shall display a small button with a trash can icon.
- REQ-36: Upon users clicking that trash can button, the system shall remove that item from the user's cart.

3.7 Ratings Section (Low Priority)

3.7.1 Description and Priority

This feature allows user to rate books after checking them out. Ratings will be on a scale of 1 to 5 stars, and users will be able to give reviews based on their ratings. Other users will be able to see the ratings books are given.

3.7.2 Stimulus/Response Sequences

- Stimulus (1): User clicks on review button for a certain book.
- Response (1): User is able to check out reviews and ratings of the book and be able to give a rating / review of the book.
- Stimulus (2): User clicks "Post" button on their review / rating.
- Response (2): Other users are able to view the rating of the book and owner of the book is notified.

- Stimulus (3): Owner of book presses own book's button to respond to ratings / reviews.
- Response (3): Owner of book is able to look at and respond to ratings / reviews.

3.7.3 Functional Requirements

- REQ-37: System shall have button available for users to look at reviews of books.
- REQ-38: System shall notify users / owners of books when review of a book is published.
- REQ-39: Inappropriate or offensive reviews are flagged if they are posted.
- REQ-40: Ratings will have 5 star buttons for users to press.
- REQ-41: Upon users clicking on a star, the corresponding number shall be added to the running total up to that point, and an updated score average shall be calculated.

3.8 Payment Section (High Priority)

3.8.1 Description and Priority

This feature is how users can check out their books, prompting users for credit card information (Simulated, no real payments will take place here for security reasons.), See how much their cart total is, and set up payment for their books.

3.8.2 Stimulus/Response Sequences

- Stimulus (1): The user clicks the Check Out button next to Cart Info.
- Response (1): Website brings user into check out page, where they can fill out info on how they are paying for the books.
- Stimulus (2): User fills out credit card information and submits payment.
- Response (2): Book is now owned by user and website prompts a "Thank You" message.
- Stimulus (3): User decides they want to get out of the check out page without paying for the books.
- Response (3): The website then removes them from the checkout page and returns them to the main shopping area.
- Stimulus (4): User has insufficient funds at the check out page.
- Response (4): Website lets user know they're unable to pay the funds required for their cart, then takes them back to the main shopping page.

3.8.3 Functional Requirements

- REQ-42: The system shall display the checkout page when a user selects the Check Out button from the cart interface.
- REQ-43: The system shall present the user with a clear summary of their cart contents, including item names, quantities, prices, and total cost before payment is submitted.
- REQ-44: The system shall provide input fields for simulated credit card information, including card number, expiration date, CVV, and billing name.
- REQ-45: The system shall validate the format of the entered payment information before allowing the user to submit the payment form.

3.9 Reporting Offensive Content (Medium Priority)

3.9.1 Description and Priority

Users that find books, usernames, and authors offensive, inappropriate, or a scam, are able to report these materials. Admin are also notified when users report a property in the website.

3.9.2 Stimulus/Response Sequences

- Stimulus (1): User sees item that is offensive or innapropriate for the website.
- Response (1): User is then able to report this item and the item is hidden from the user.
- Stimulus (2): Admin is notified of a user reporting the book or author.
- Response (2): Admin has option to remove the item that was reported or allow it to stay if there is no reason to remove it.

3.9.3 Functional Requirements

- REQ-46: The system shall provide users with a clearly visible option to report books, usernames, or authors they deem offensive, inappropriate, or fraudulent.
- REQ-47: The system shall allow users to submit a report that identifies the specific item and includes an optional explanation of why it is being reported.
- REQ-48: The system shall hide the reported item from the reporting user's view immediately after the report is submitted.
- REQ-49: The system shall notify administrators whenever a user submits a report regarding a book, username, or author.

3.10 Book Comparison Tool (Low Priority)

3.10.1 Description and Priority

This feature allows users to compare multiple listings of the same book side-by-side based on price, condition, seller rating, and shipping cost.

3.10.2 Stimulus/Response Sequences

- Stimulus (1): User searches for a specific book.
- Response (1): System displays multiple listings for the same book from different sellers.
- Stimulus (2): User selects "Compare" option.
- Response (2): System displays a comparison table showing price, condition, seller rating, and total cost for each listing.
- Stimulus (3): User selects preferred listing from comparison view.
- Response (3): System adds selected item to cart or navigates to product detail page.

3.10.3 Functional Requirements

- REQ-50: The system shall group multiple listings of the same book (matched by ISBN) in search results.
- REQ-51: The system shall display a comparison view showing price, condition, seller rating, shipping cost, and total cost.
- REQ-52: The system shall highlight the best value option based on total cost and condition.
- REQ-53: The system shall allow sorting comparison results by price, condition, or seller rating.

3.11 Wishlist (Low Priority)

3.11.1 Description and Priority

This feature allows users to save books they are interested in for future reference without adding them to their cart immediately.

3.11.2 Stimulus/Response Sequences

- Stimulus (1): User clicks "Add to Wishlist" button on a product page.
- Response (1): System adds the book to the user's Wishlist and displays confirmation.
- Stimulus (2): User navigates to their Wishlist page.
- Response (2): System displays all saved books with current prices and availability status.
- Stimulus (3): User removes item from Wishlist.
- Response (3): System removes the item and updates the Wishlist display.

3.11.3 Functional Requirements

- REQ-54: The system shall provide an "Add to Wishlist" button on each product listing page.
- REQ-55: The system shall display a dedicated Wishlist page accessible from the user's account.
- REQ-56: The system shall show updated pricing and availability for Wishlisted items.
- REQ-57: The system shall allow users to move items directly from Wishlist to cart.
- REQ-58: The system shall automatically remove items from Wishlists if they are deleted by the seller.

3.12 Orders Page (High Priority)

3.12.1 Description and Priority

This feature provides users with the ability to see their order history, click on a specific order for more details, and initiate a return.

3.12.2 Stimulus/Response Sequences

- Stimulus (1): User access orders page.
- Response (1): System shall show a canonical log of all orders that a user has placed.
- Stimulus (2): User clicks on a specific order.
- Response (2): System shall focus on that page and provide more details than on the canonical logs page.

- Stimulus (3): User clicks on "initiate a return".
- Response (3): System shall change order status to "Returned".

3.12.3 Functional Requirements

- REQ-59: The system shall display all orders for a user in reverse chronological order.
- REQ-60: All orders shall be clickable, and the system shall provide more information about an order once it is clicked.
- REQ-61: The system shall clearly display a "return" button.
- REQ-62: The system shall change an order's status to "RETURNED" after the user clicks on the "return" button and confirms their action.

3.13 Add a book page (High Priority)

3.13.1 Description and Priority

This feature provides users with the ability to publish books for sale.

3.13.2 Stimulus/Response Sequences

- Stimulus (1): User types into the "Book Title" text field.
- Response (1): System saves the input to use in the listing.
- Stimulus (2): User types into the "Author" text field.
- Response (2): System saves the input to use in the listing.
- Stimulus (3): User types into the "ISBN" text field.
- Response (3): System saves the input to use in the listing.
- Stimulus (4): User types into the "Genre" text field.
- Response (4): System saves the input to use in the listing.
- Stimulus (5): User types into the "Edition" text field.
- Response (5): System saves the input to use in the listing.
- Stimulus (6): User types into the "Year" text field.
- Response (6): System saves the input to use in the listing.
- Stimulus (7): User types into the "Price" number field.

- Response (7): System saves the input to use in the listing.
- Stimulus (8): User types or uses the arrows to modify the "In Stock" number field.
- Response (8): System saves the input to use in the listing.
- Stimulus (9): User selects an option from the "Condition" drop-down menu.
- Response (9): System saves the input to use in the listing.
- Stimulus (10): User types into the "Description" text field.
- Response (10): System saves the input to use in the listing.
- Stimulus (11): User clicks on the "Post Listing" button with all previous fields containing valid inputs.
- Response (11): System creates a database entry and puts listing in a queue to be authorized by an admin.
- Stimulus (12): User clicks on the "Post Listing" button with any field containing invalid inputs.
- Response (12): System notifies the user of this and asks them to correct the information.

3.13.3 Functional Requirements

- REQ-59: The system shall display text boxes for the "Book Title", "Author", "ISBN Number", "Genre", "Edition" and "Description" fields.
- REQ-60: The system shall display number boxes for the "Year", "Price" and "In Stock" fields.
- REQ-61: The system shall display a "Condition" drop-down menu with the options "NEW", "LIKE NEW", "VERY GOOD", "GOOD" and "ACCEPTABLE".
- REQ-62: The system shall display a "Post Listing" button, which shall create a listing if all entry fields are valid.

3.14 Filtering and Sorting (Low Priority)

3.14.1 Description and Priority

This feature provides users with additional filtering options to optimize search results by multiple criteria including genre, condition, seller rating, and location.

3.14.2 Stimulus/Response Sequences

- Stimulus (1): User selects multiple filter criteria (genre, condition, seller rating).
- Response (1): System applies all filters and displays matching results.
- Stimulus (2): User adjusts filter settings.
- Response (2): System dynamically updates results in real-time.
- Stimulus (3): User saves filter preferences.
- Response (3): System stores preferences for future searches.

3.14.3 Functional Requirements

- REQ-63: The system shall allow filtering by genre with multi-select checkboxes.
- REQ-64: The system shall provide a condition filter with options for New, Like New, Very Good, Good, and Acceptable.
- REQ-65: The system shall allow filtering by minimum seller rating (1-5 stars).
- REQ-66: The system shall provide location-based filtering to show nearby sellers.

4 Other Nonfunctional Requirements

4.1 Performance Requirements

- The system will provide responsive performance for all core user interactions, including viewing listings, browsing categories, and performing click-based actions. Under normal operating conditions, these actions should be completed within an industry-standard response time of no more than 2 seconds.
- The system will support concurrent access to listings, allowing a single listing to be added to multiple users' shopping carts simultaneously. When a user completes the checkout process for a listing, the system will synchronize in real time to ensure that the listing is immediately removed from all other active carts.
- This synchronization mechanism will also apply when a listing is removed or when a user or moderator deletes an account. In those cases, all associated listings and cart references will be updated across the system within 2 seconds to maintain data consistency and prevent invalid transactions.

4.2 Safety Requirements

- The website will not endorse user-to-user transactions; the company that owns the website will always act as an intermediary to ensure both sides of the transaction hold up their end and to resolve any possible disputes. For this end, chats between users will not be implemented.
- The system will comply with PCI DSS (Payment Card Industry Data Security Standard) requirements for handling payment-related info through approved third party payment processors.
- The system shall implement automated fraud detection algorithms that flag suspicious transactions based on patterns such as unusually high-value purchases, rapid successive transactions, or mismatched billing/shipping information.
- The system shall maintain uptime of at least 99.5% to ensure continuous service availability for users.

4.3 Security Requirements

- Some user information will be visible to other users, including username and real name. This is done for transparency, which users will appreciate. Other sensible

information, such as address and email, will be kept private from other users and will be encrypted in the database to prevent any possible hackers from accessing this information.

- All transactions will be handled with industry-standard privacy and security measures in mind, such as validating and encrypting card information.
- Users' actions will be able to be moderated. This includes censoring of potential offensive language, censoring of private data placed on public fields, etc.
- All user passwords will be stored using hashing algorithms.
- Communication between client and server will occur on *HTTPS*.
- System will log out inactive users after an agreed upon timeout period.
- System must maintain audit logs for financial transactions and administrative actions.
- Passwords will have to be at least 10 letters with at least one letter, number, and punctuation symbol.

4.4 Software Quality Attributes

Usability:

- The system shall provide an intuitive user interface that requires no more than 3 clicks to complete any core function (search, add to cart, checkout).
- The system shall maintain consistent navigation and layout across all pages to reduce learning curve.
- The system shall provide clear error messages that explain what went wrong and how to fix it, avoiding technical jargon.
- The system shall be accessible to users with disabilities, complying with WCAG 2.1 Level AA standards, including screen reader compatibility and keyboard navigation.

Reliability:

- The system shall maintain 99.5% uptime during normal operating conditions.
- The system shall handle unexpected errors gracefully without crashing, displaying user-friendly error pages instead of stack traces.
- The system shall implement automatic recovery mechanisms for database connection failures.

Maintainability:

- The system shall include comprehensive code documentation and inline comments to facilitate future maintenance.
- The system shall use version control (Git) to track all code changes and enable rollback if necessary.
- The system shall maintain a separate development, staging, and production environment to test updates before deployment.

Scalability:

- The system architecture shall support horizontal scaling to handle increased user traffic during peak periods.
- The database shall be designed to efficiently handle up to 100,000 active listings and 50,000 registered users without performance degradation.
- The system shall support adding new features and modules without requiring complete system redesign.

Portability:

- The system shall be compatible with major web browsers including Chrome, Firefox, Safari, and Edge (latest two versions of each).
- The system shall provide a responsive design that adapts to desktop, tablet, and mobile screen sizes.
- The system shall be deployable on standard web hosting platforms without requiring proprietary infrastructure.

Testability:

- The system shall include automated unit tests covering at least 80% of backend code.
- The system shall implement integration tests for all critical user workflows (registration, search, checkout).
- The system shall provide logging mechanisms that record errors, transactions, and user actions for debugging purposes.

Performance:

- The system shall load the homepage within 2 seconds under normal network conditions.
- The system shall process search queries and return results within 1 second for databases up to 100,000 listings.
- The system shall complete checkout transactions within 3 seconds of payment submission.

5 Other Requirements

5.1 Database Requirements

- The system shall use PostgreSQL as a relational database management system to store all application data.
- The system shall implement proper database indexing on frequently queried fields (user_id, book_id, ISBN) to optimize search performance.
- The system shall enforce referential integrity through foreign key constraints to maintain data consistency across related tables.

5.2 Legal and Regulatory Requirements

- The system shall comply with the Children’s Online Privacy Protection Act (COPPA) by requiring users to be at least 13 years old to create an account.
- The system shall comply with applicable state and federal consumer protection laws regarding online sales and refund policies.
- The system shall display a clear Terms of Service and Privacy Policy that users must accept before creating an account.
- The system shall comply with accessibility requirements under Section 508 of the Rehabilitation Act and the Americans with Disabilities Act (ADA).

5.3 Internationalization Requirements

- English will be the only language for this website.
- The system shall display prices in USD (\$) with proper formatting (e.g., \$19.99).
- The system shall support US date format (MM/DD/YYYY) in the initial release with the ability to support alternative formats in future versions.